

Spiking Neural Network (SNN) Robustness Benchmark
Mixed Comparison Report: CNN Baseline vs. SNN (Atan
 $\alpha \in \{1.2, 2.0\}$ and Tri $\beta = 2.0$)

ByzFL SNN Benchmark

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1 Introduction and Experimental Settings

This report provides a side-by-side mixed comparison of four distinct configurations evaluated under identical federated learning setups featuring varying data heterogeneity (non-IID levels) and Byzantine attacks:

1. **CNN Baseline**: A standard (non-spiking) Convolutional Neural Network (`cnn_mnist`).
2. **SNN Atan** ($\alpha = 1.2$): A Spiking CNN (`cnn_mnist_snn`) training with the ArcTangent surrogate gradient at stiffness $\alpha = 1.2$.
3. **SNN Atan** ($\alpha = 2.0$): A Spiking CNN (`cnn_mnist_snn`) training with the ArcTangent surrogate gradient at stiffness $\alpha = 2.0$.
4. **SNN Tri** ($\beta = 2.0$): A Spiking CNN (`cnn_mnist_snn`) training with the Triangular surrogate gradient at parameter $\beta = 2.0$.

These configurations allow direct comparative analysis of the non-spiking model performance against optimized Spiking Neural Networks using different surrogate gradient functions and parameters.

1.1 Parameter Configurations

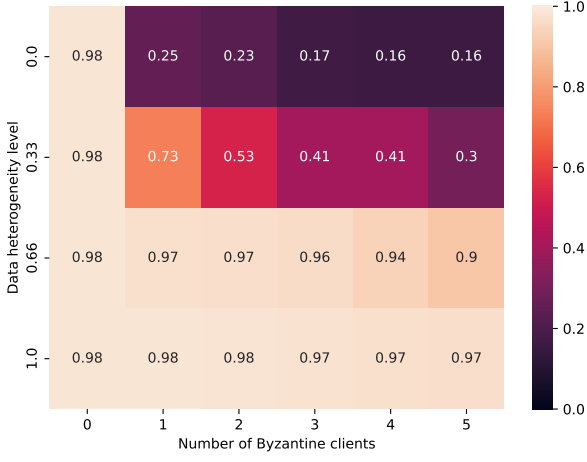
Table 1 details the parameters, hyperparameters, and algorithms used in this comparison.

Table 1: Experiment Parameters and Hyperparameters

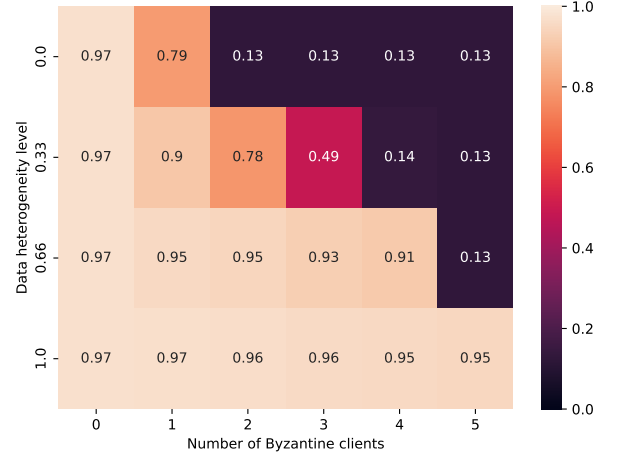
Component / Parameter	Configuration Value
Dataset	MNIST (60,000 train, 10,000 test)
Model Structure (CNN)	Convolutional Neural Network (<code>cnn_mnist</code>)
Model Structure (SNN)	Spiking Convolutional Neural Network (<code>cnn_mnist_snn</code>)
Loss Function	Rate-coded Cross-Entropy Loss (<code>ce_rate_loss</code>)
Optimizer	SGD
Learning Rate	0.10
Momentum	0.9
Weight Decay	0.0001
Batch Size	128
SNN Encoding	Constant encoding, 10 time steps
SNN Decay (β)	0.95
SNN Threshold	1.0 (fixed)
Training Steps	500
Training Algorithm	DSGD
Honest Nodes (N)	10
Byzantine Nodes (f)	$f \in \{0, 1, 2, 3, 4, 5\}$
Data Heterogeneity (γ)	$\gamma \in \{1.0, 0.66, 0.33, 0.0\}$ (from IID to extreme Non-IID)
Pre-aggregators	NNM, ARC
Aggregators Swept	Centered Clipping, Geometric Median, Multi-Krum, Trimmed Mean
Attacks Evaluated	Optimal ALIE, Sign Flipping, Optimal Inner Product Manipulation
Training Seeds	5 seeds (42 to 46)
Data Dist. Seed	42

2 Best Overall Performance (Worst-Case Across Attacks)

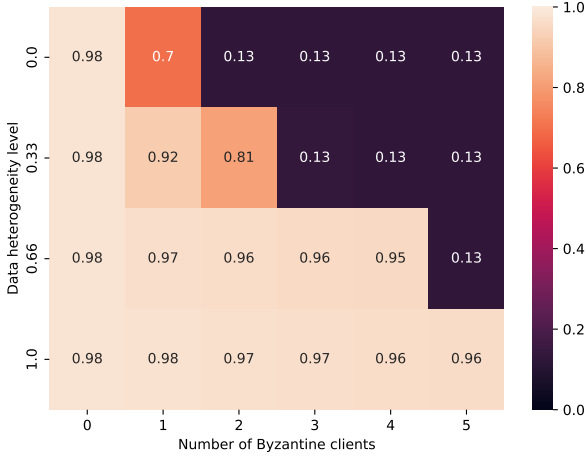
The heatmaps in this section display the worst-case test accuracy at convergence across all attacks achieved by selecting the best robust aggregator for each cell.



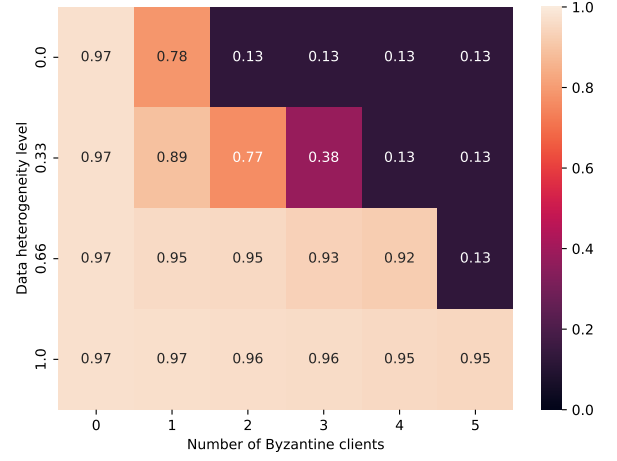
(a) CNN Baseline



(b) SNN Atan ($\alpha = 1.2$)



(c) SNN Atan ($\alpha = 2.0$)



(d) SNN Tri ($\beta = 2.0$)

Figure 1: Best overall test accuracy heatmaps (worst-case across attacks) for CNN Baseline vs. SNN Atan ($\alpha = 1.2, 2.0$) vs. SNN Tri ($\beta = 2.0$).

3 Best Aggregator under Specific Attacks

This section presents heatmaps showing the best aggregator performance under each specific attack.

3.1 Optimal A Little Is Enough (ALIE) Attack

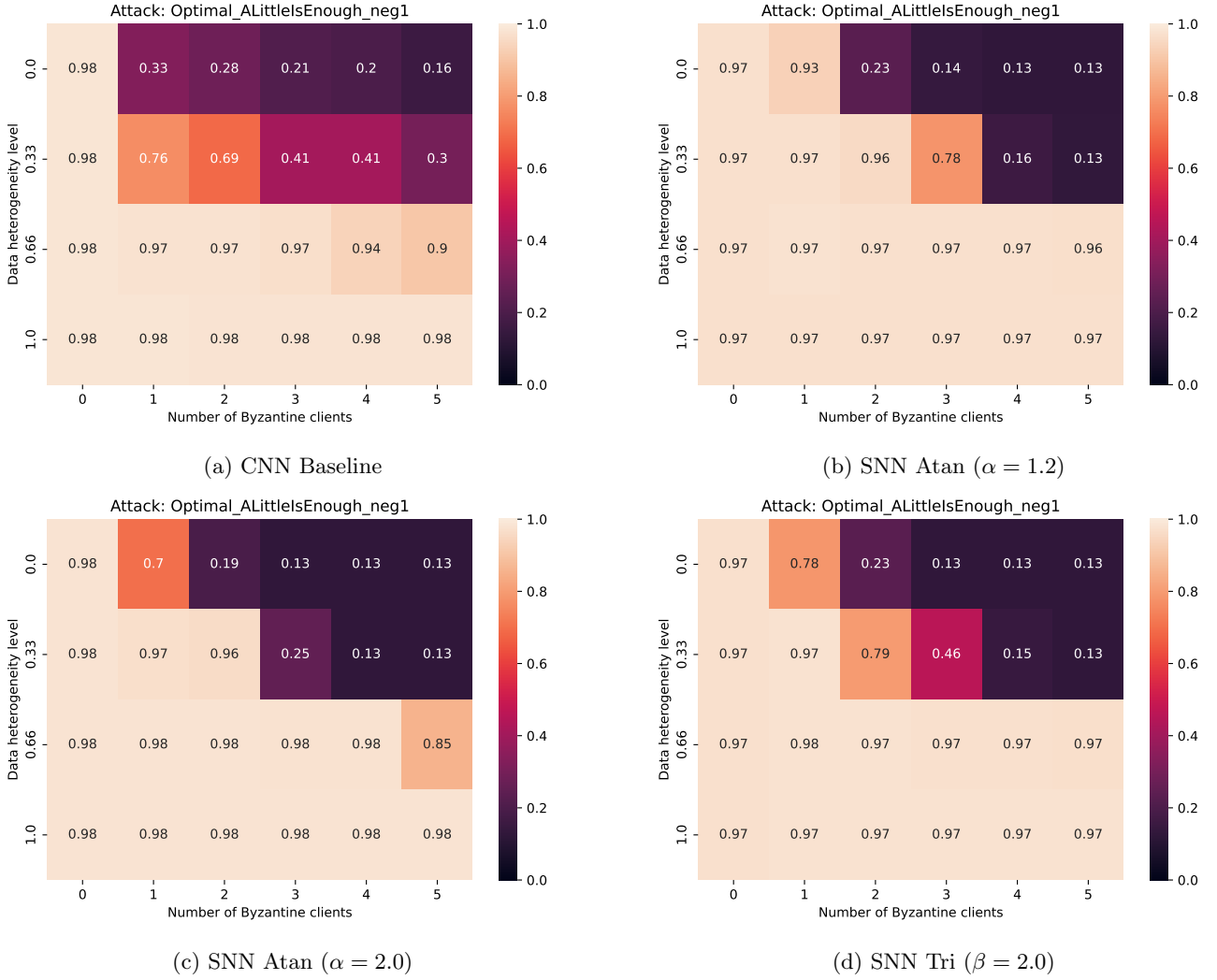
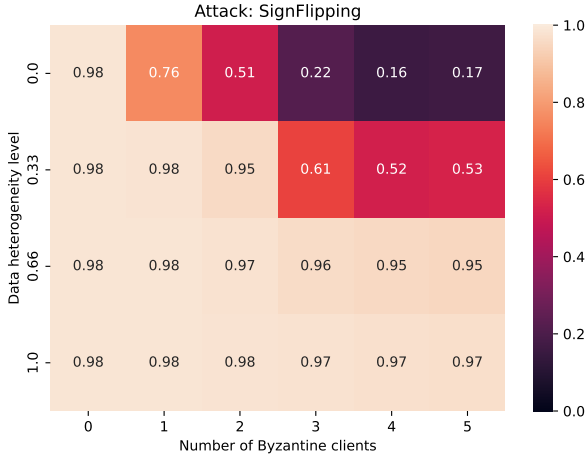


Figure 2: Best test accuracy heatmaps under the Optimal ALIE attack for CNN Baseline vs. SNN Atan ($\alpha = 1.2$, 2.0) vs. SNN Tri ($\beta = 2.0$).

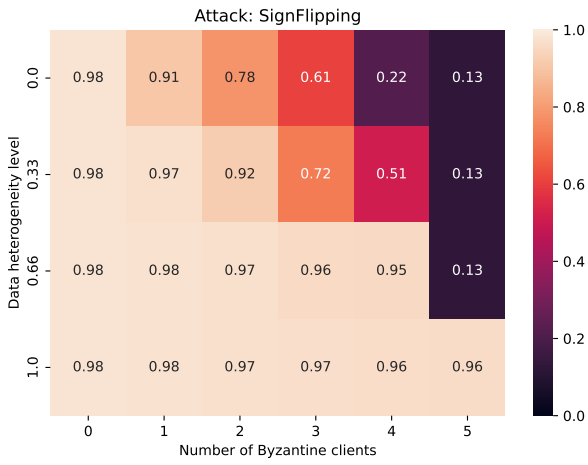
3.2 Sign Flipping Attack



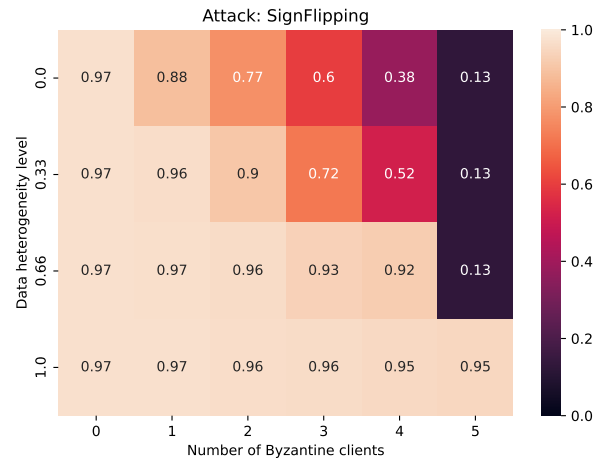
(a) CNN Baseline



(b) SNN Atan ($\alpha = 1.2$)



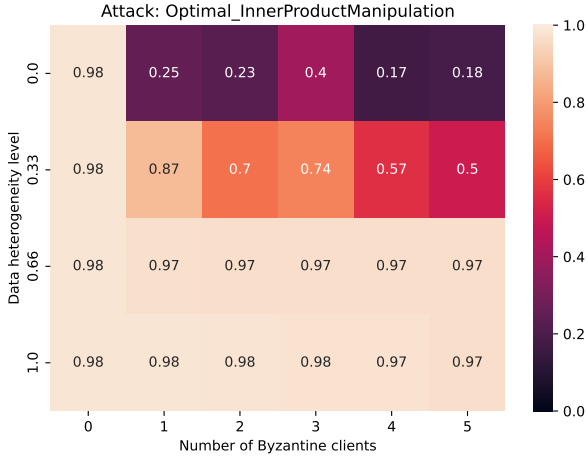
(c) SNN Atan ($\alpha = 2.0$)



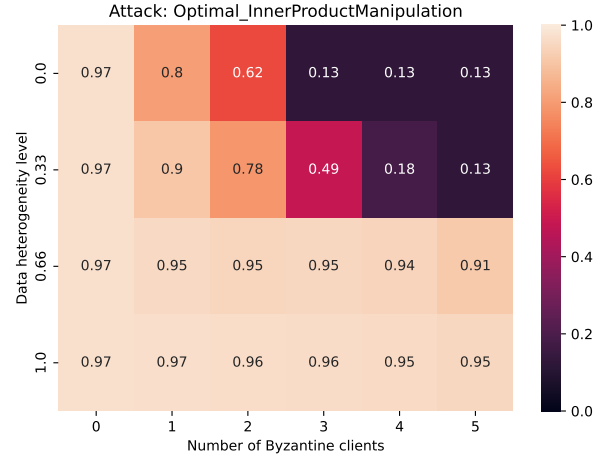
(d) SNN Tri ($\beta = 2.0$)

Figure 3: Best test accuracy heatmaps under the Sign Flipping attack for CNN Baseline vs. SNN Atan ($\alpha = 1.2$, 2.0) vs. SNN Tri ($\beta = 2.0$).

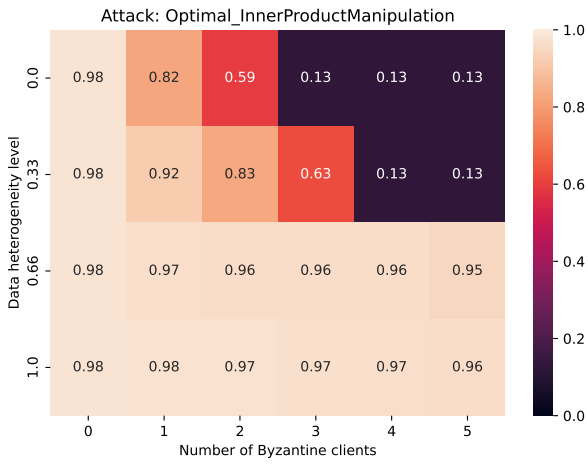
3.3 Optimal Inner Product Manipulation Attack



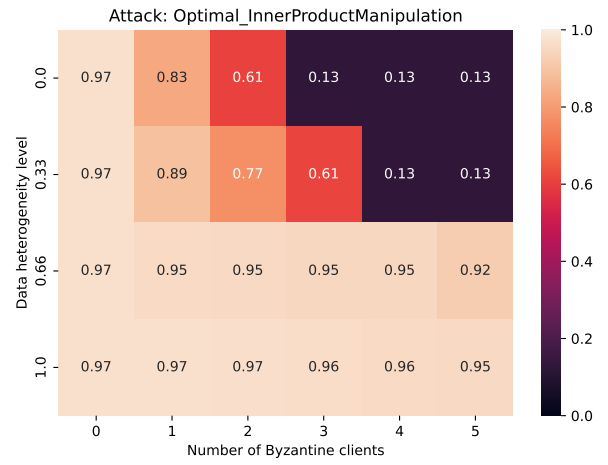
(a) CNN Baseline



(b) SNN Atan ($\alpha = 1.2$)



(c) SNN Atan ($\alpha = 2.0$)



(d) SNN Tri ($\beta = 2.0$)

Figure 4: Best test accuracy heatmaps under the Optimal Inner Product Manipulation attack for CNN Baseline vs. SNN Atan ($\alpha = 1.2, 2.0$) vs. SNN Tri ($\beta = 2.0$).

4 Robustness of Fixed Aggregators across all Attacks

In this section, we analyze the performance of each specific aggregator (Centered Clipping, Geometric Median, Multi-Krum, Trimmed Mean) across all Byzantine attack settings.

4.1 Centered Clipping (CC)

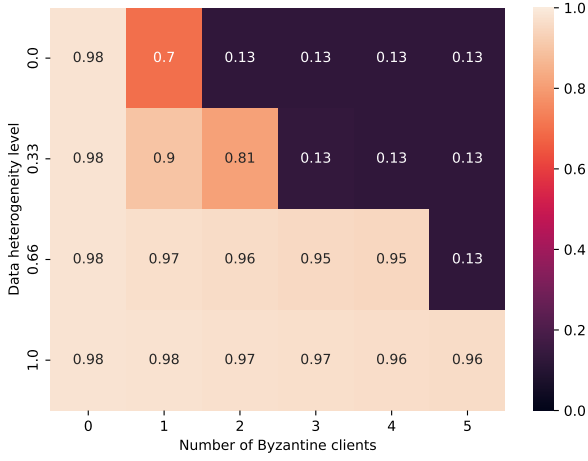
4.1.1 CC under Worst-Case across all Attacks



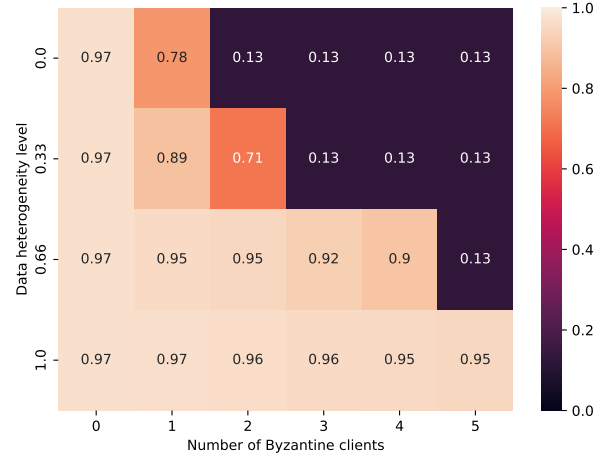
(a) CNN Baseline



(b) SNN Atan ($\alpha = 1.2$)



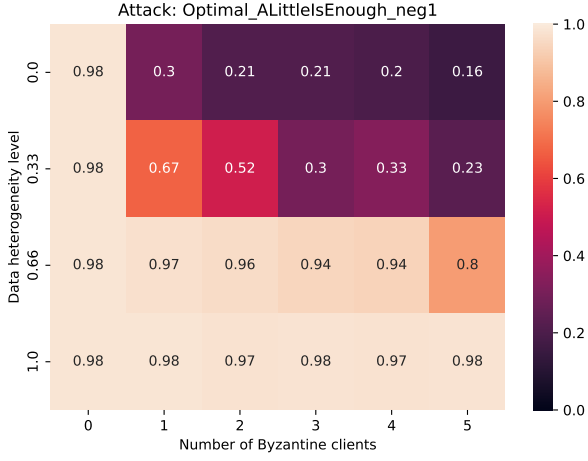
(c) SNN Atan ($\alpha = 2.0$)



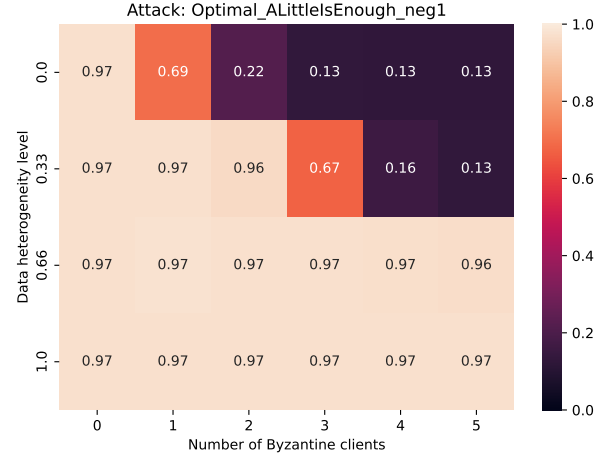
(d) SNN Tri ($\beta = 2.0$)

Figure 5: Centered Clipping performance under the worst-case across all attacks for CNN Baseline vs. SNN Atan ($\alpha = 1.2, 2.0$) vs. SNN Tri ($\beta = 2.0$).

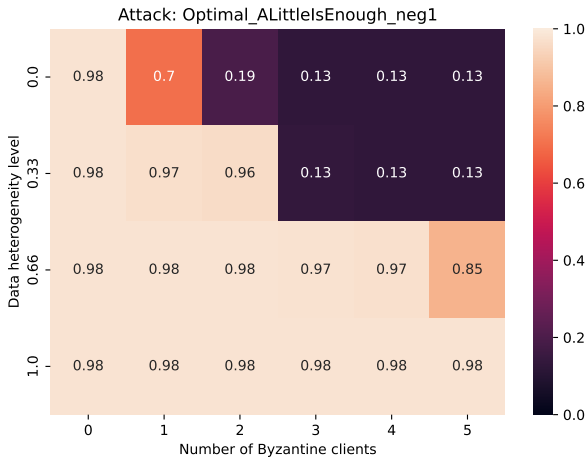
4.1.2 CC under Optimal ALIE Attack



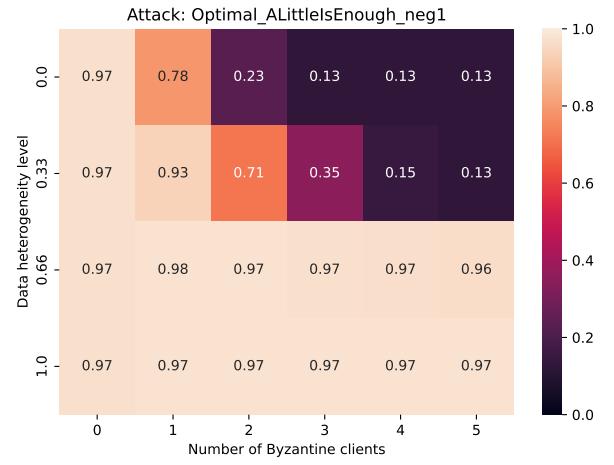
(a) CNN Baseline



(b) SNN Atan ($\alpha = 1.2$)



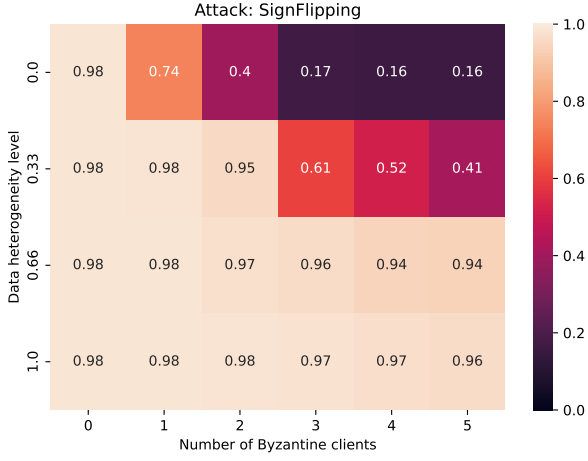
(c) SNN Atan ($\alpha = 2.0$)



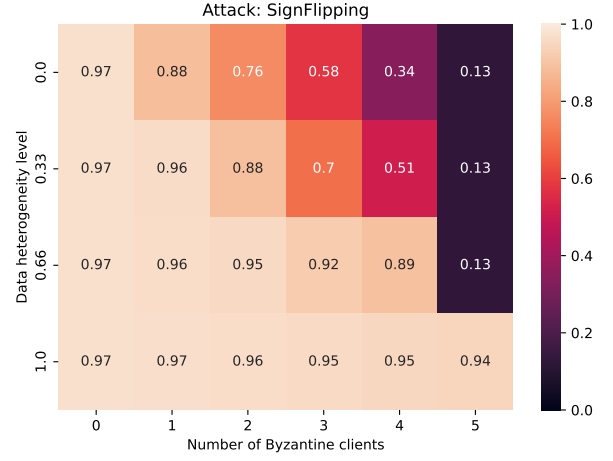
(d) SNN Tri ($\beta = 2.0$)

Figure 6: Centered Clipping performance under Optimal ALIE for CNN Baseline vs. SNN Atan ($\alpha = 1.2, 2.0$) vs. SNN Tri ($\beta = 2.0$).

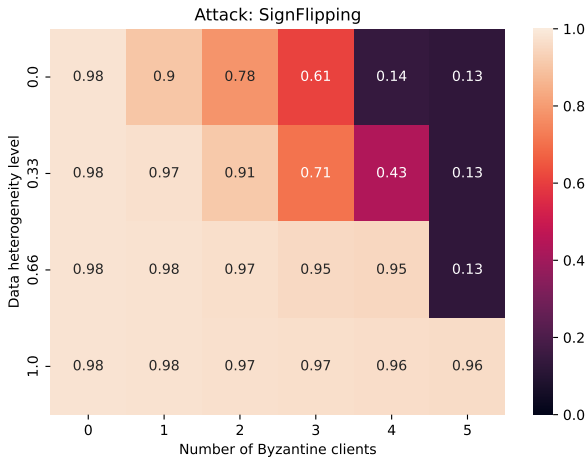
4.1.3 CC under Sign Flipping Attack



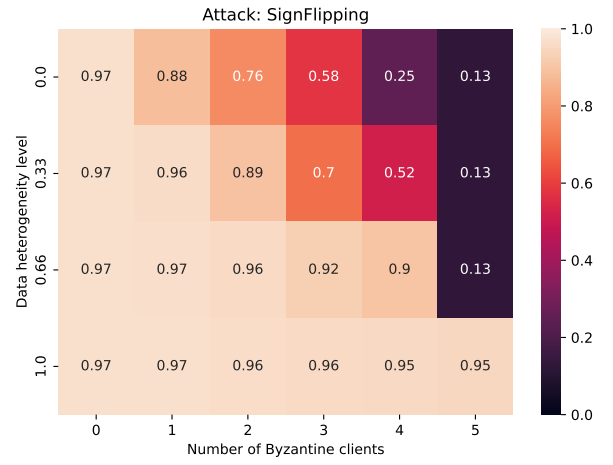
(a) CNN Baseline



(b) SNN Atan ($\alpha = 1.2$)



(c) SNN Atan ($\alpha = 2.0$)



(d) SNN Tri ($\beta = 2.0$)

Figure 7: Centered Clipping performance under Sign Flipping for CNN Baseline vs. SNN Atan ($\alpha = 1.2, 2.0$) vs. SNN Tri ($\beta = 2.0$).

4.1.4 CC under Optimal Inner Product Manipulation Attack

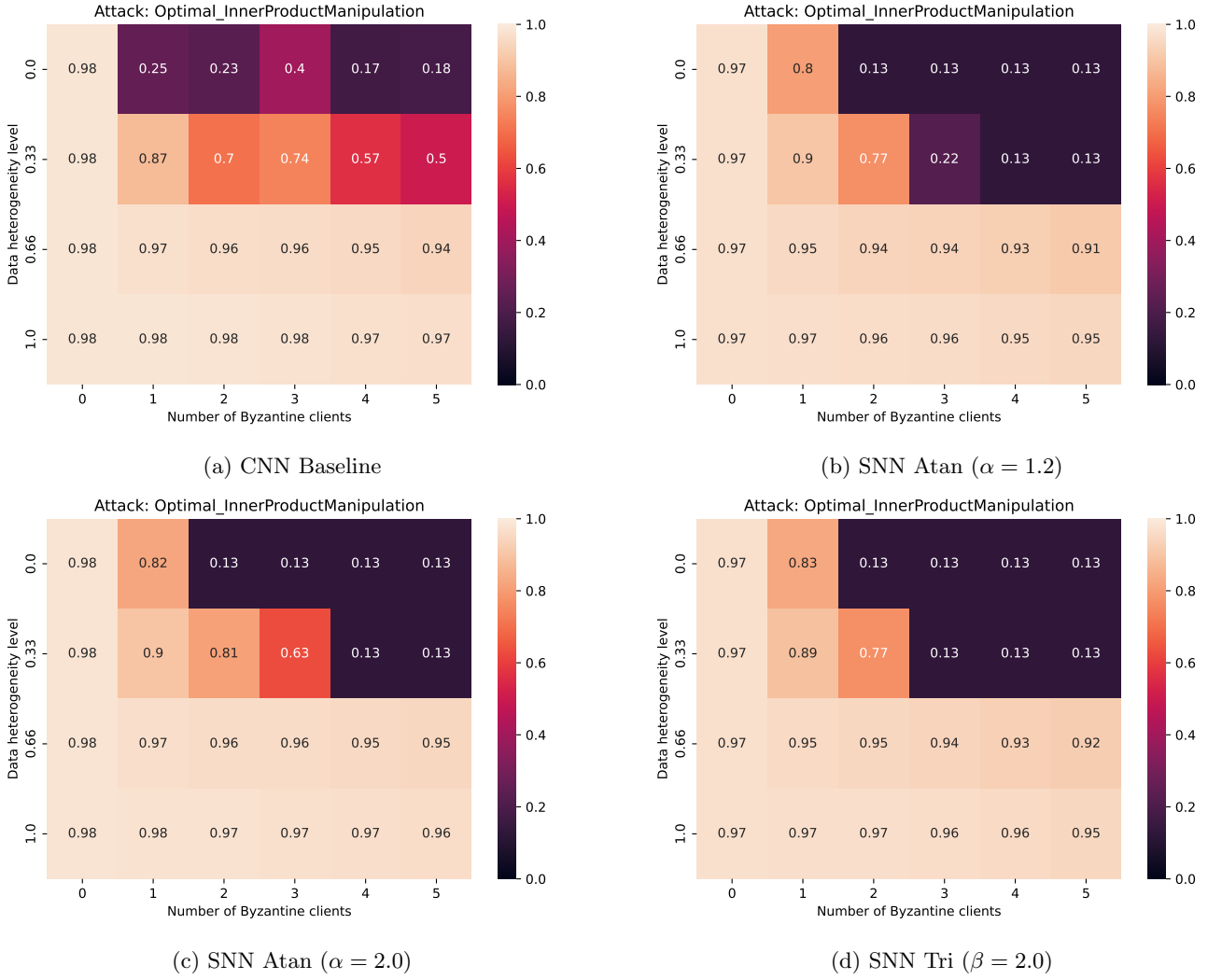
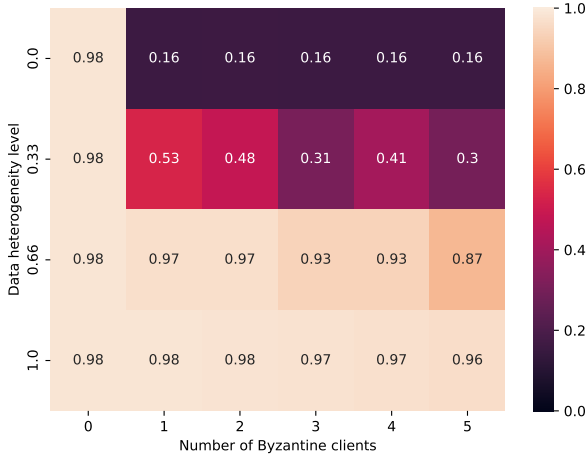


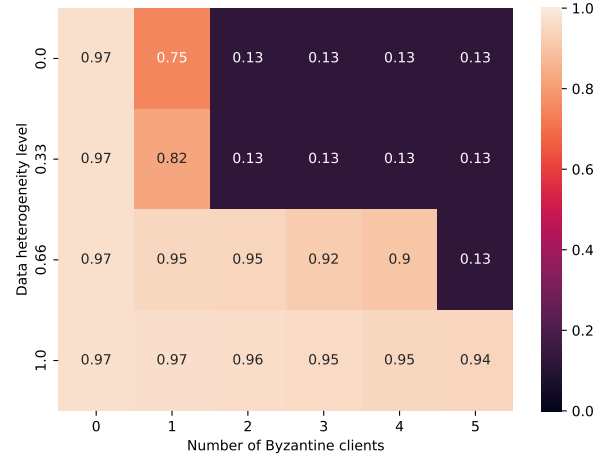
Figure 8: Centered Clipping performance under Optimal Inner Product Manipulation for CNN Baseline vs. SNN Atan ($\alpha = 1.2, 2.0$) vs. SNN Tri ($\beta = 2.0$).

4.2 Geometric Median (GM)

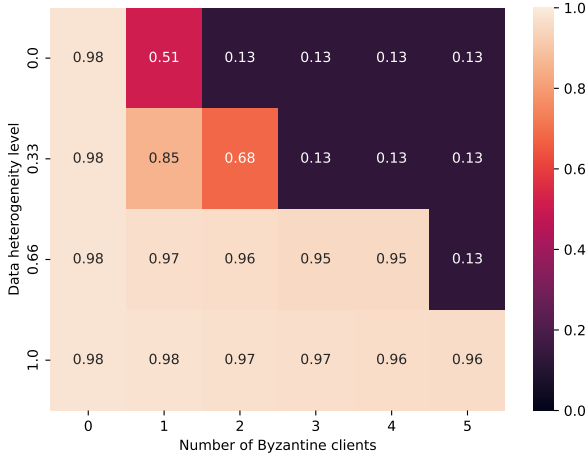
4.2.1 GM under Worst-Case across all Attacks



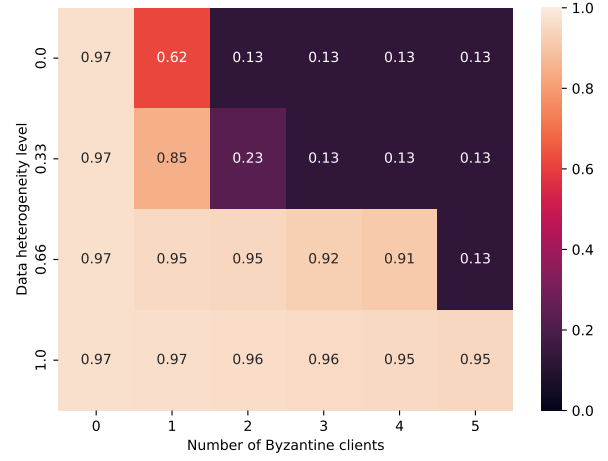
(a) CNN Baseline



(b) SNN Atan ($\alpha = 1.2$)



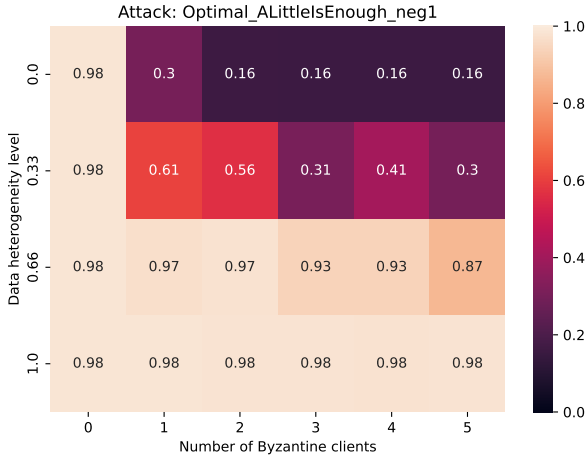
(c) SNN Atan ($\alpha = 2.0$)



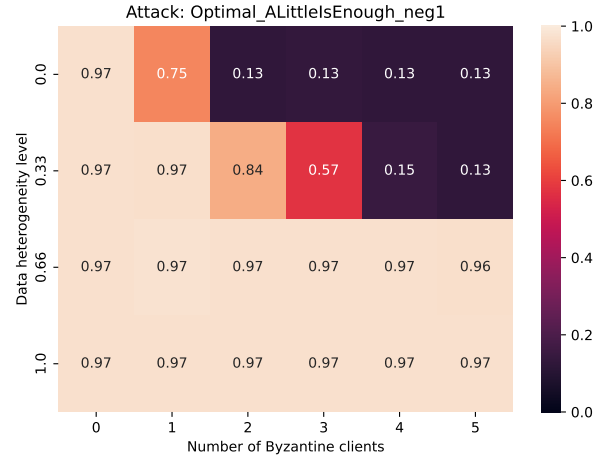
(d) SNN Tri ($\beta = 2.0$)

Figure 9: Geometric Median performance under the worst-case across all attacks for CNN Baseline vs. SNN Atan ($\alpha = 1.2, 2.0$) vs. SNN Tri ($\beta = 2.0$).

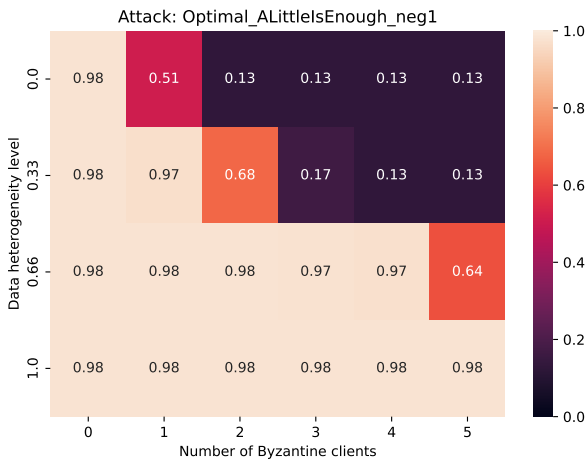
4.2.2 GM under Optimal ALIE Attack



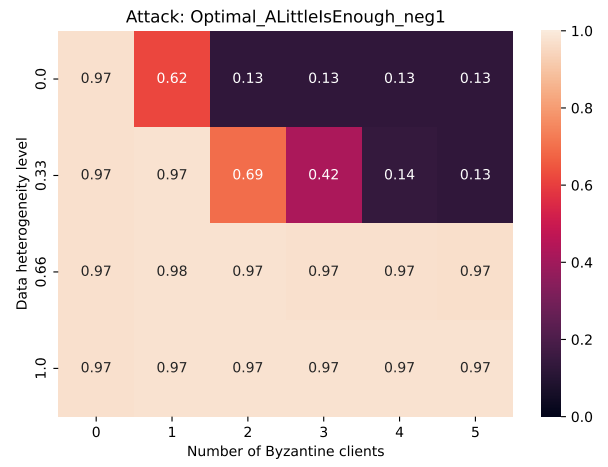
(a) CNN Baseline



(b) SNN Atan ($\alpha = 1.2$)



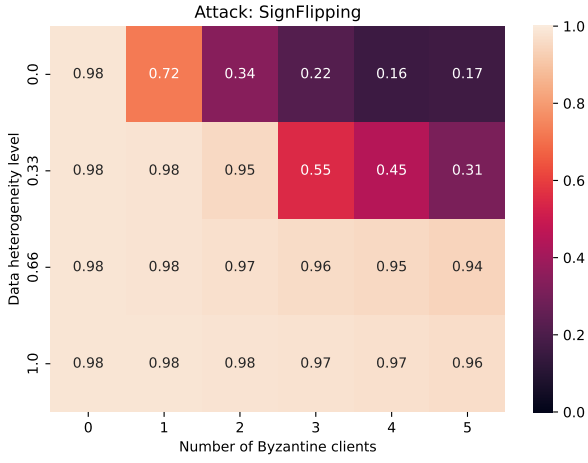
(c) SNN Atan ($\alpha = 2.0$)



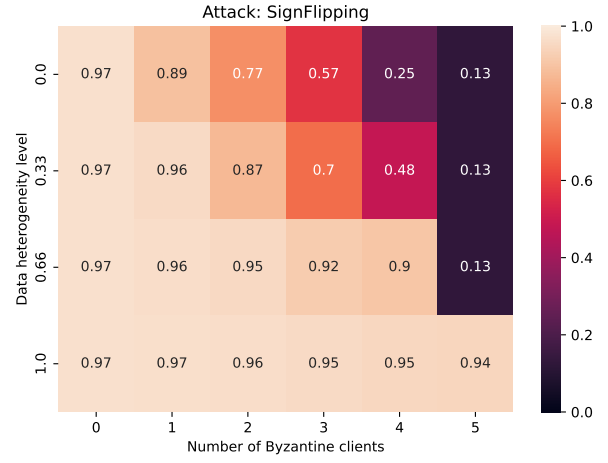
(d) SNN Tri ($\beta = 2.0$)

Figure 10: Geometric Median performance under Optimal ALIE for CNN Baseline vs. SNN Atan ($\alpha = 1.2, 2.0$) vs. SNN Tri ($\beta = 2.0$).

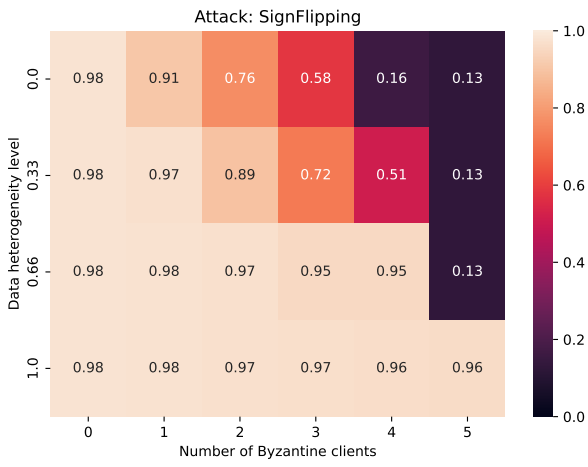
4.2.3 GM under Sign Flipping Attack



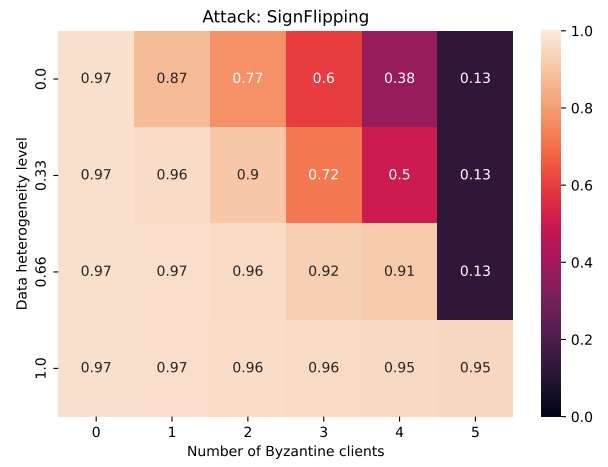
(a) CNN Baseline



(b) SNN Atan ($\alpha = 1.2$)



(c) SNN Atan ($\alpha = 2.0$)



(d) SNN Tri ($\beta = 2.0$)

Figure 11: Geometric Median performance under Sign Flipping for CNN Baseline vs. SNN Atan ($\alpha = 1.2, 2.0$) vs. SNN Tri ($\beta = 2.0$).

4.2.4 GM under Optimal Inner Product Manipulation Attack

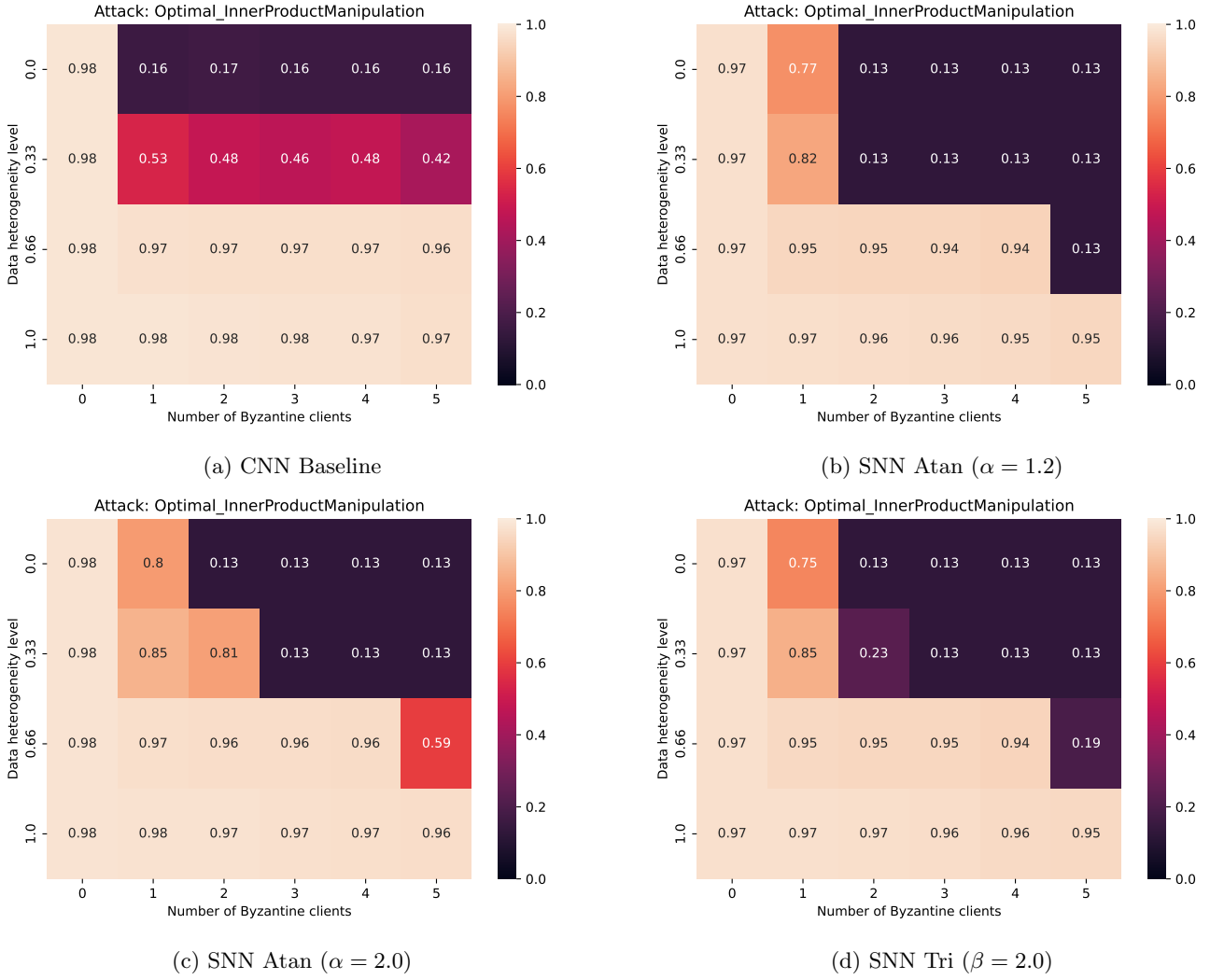
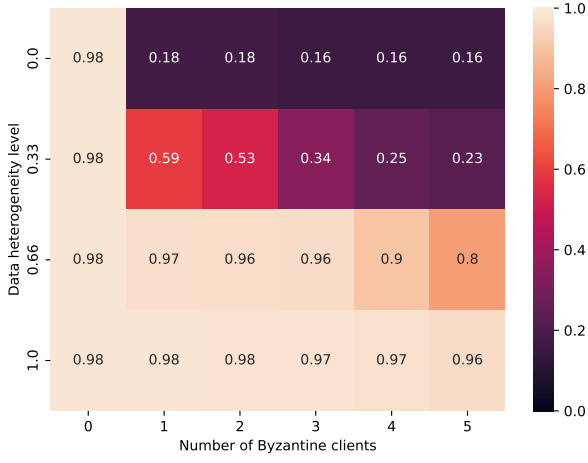


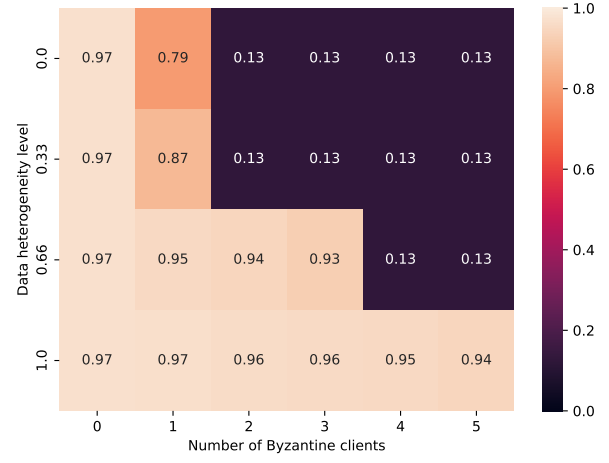
Figure 12: Geometric Median performance under Optimal Inner Product Manipulation for CNN Baseline vs. SNN Atan ($\alpha = 1.2, 2.0$) vs. SNN Tri ($\beta = 2.0$).

4.3 Multi-Krum (MK)

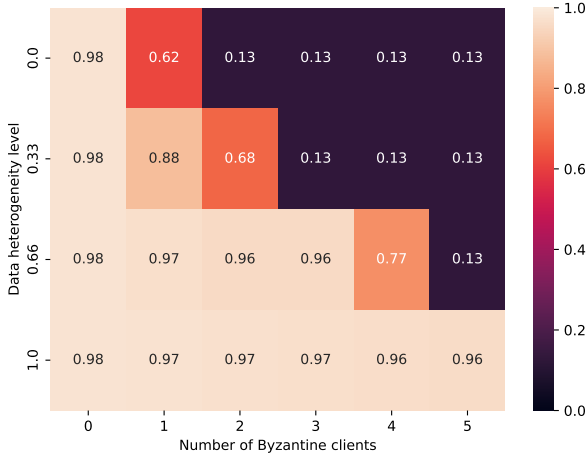
4.3.1 MK under Worst-Case across all Attacks



(a) CNN Baseline



(b) SNN Atan ($\alpha = 1.2$)



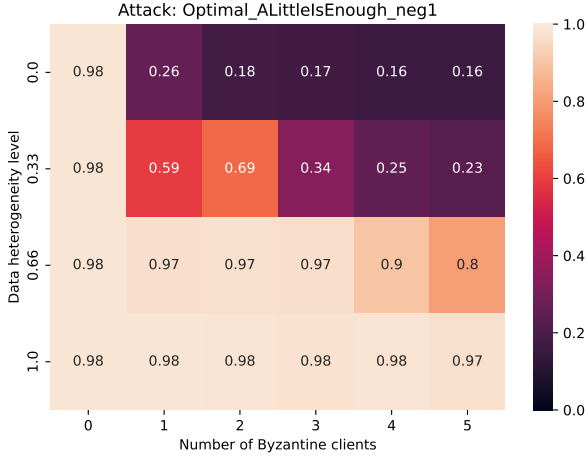
(c) SNN Atan ($\alpha = 2.0$)



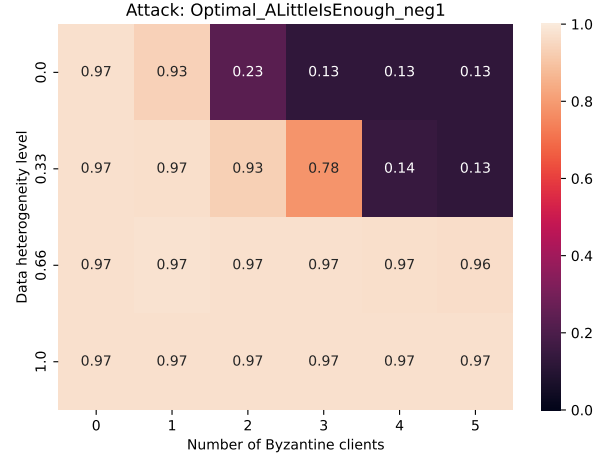
(d) SNN Tri ($\beta = 2.0$)

Figure 13: Multi-Krum performance under the worst-case across all attacks for CNN Baseline vs. SNN Atan ($\alpha = 1.2, 2.0$) vs. SNN Tri ($\beta = 2.0$).

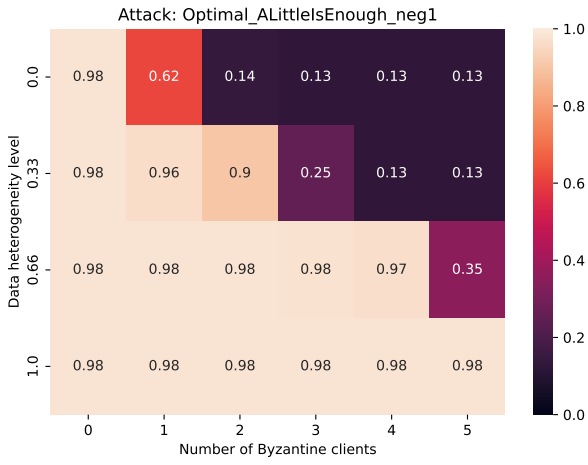
4.3.2 MK under Optimal ALIE Attack



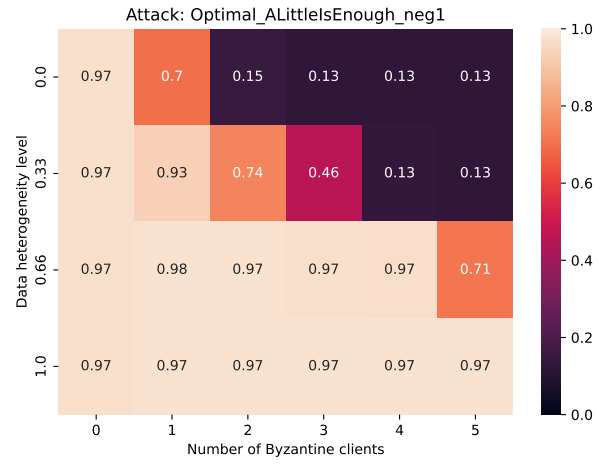
(a) CNN Baseline



(b) SNN Atan ($\alpha = 1.2$)



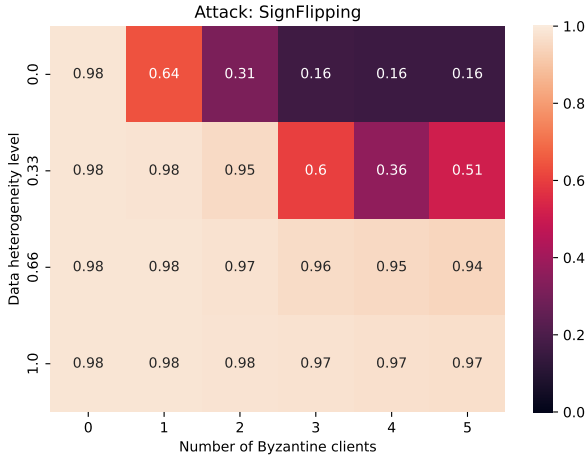
(c) SNN Atan ($\alpha = 2.0$)



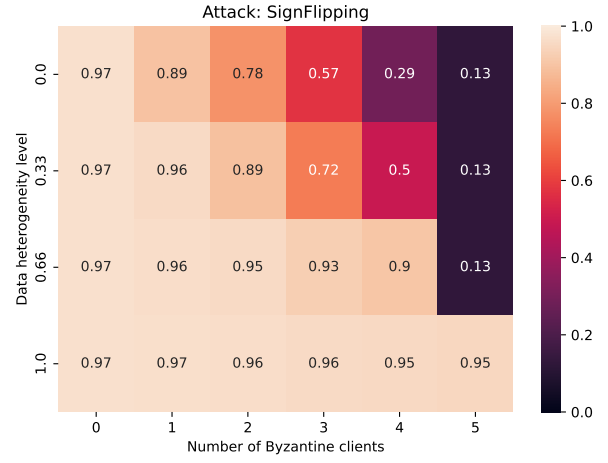
(d) SNN Tri ($\beta = 2.0$)

Figure 14: Multi-Krum performance under Optimal ALIE for CNN Baseline vs. SNN Atan ($\alpha = 1.2, 2.0$) vs. SNN Tri ($\beta = 2.0$).

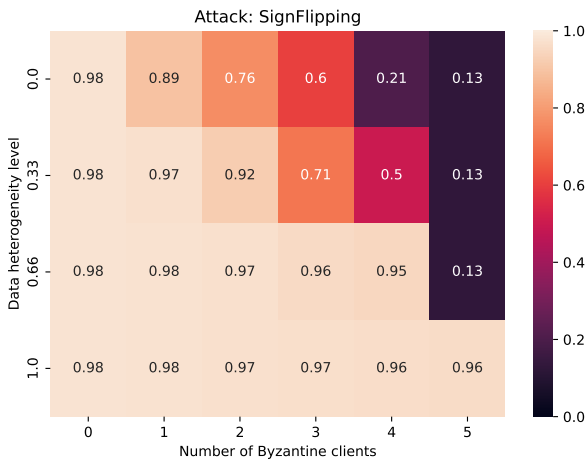
4.3.3 MK under Sign Flipping Attack



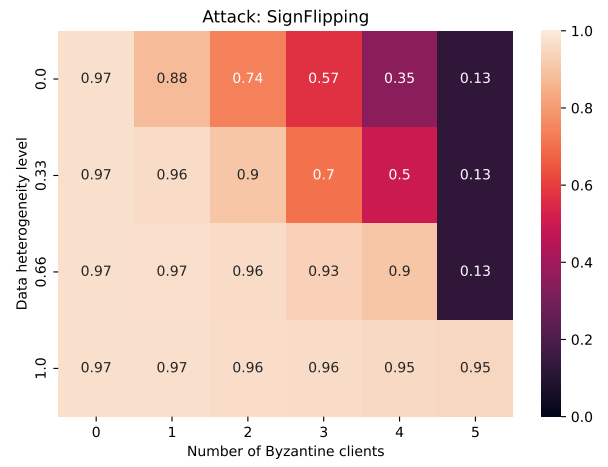
(a) CNN Baseline



(b) SNN Atan ($\alpha = 1.2$)



(c) SNN Atan ($\alpha = 2.0$)



(d) SNN Tri ($\beta = 2.0$)

Figure 15: Multi-Krum performance under Sign Flipping for CNN Baseline vs. SNN Atan ($\alpha = 1.2, 2.0$) vs. SNN Tri ($\beta = 2.0$).

4.3.4 MK under Optimal Inner Product Manipulation Attack

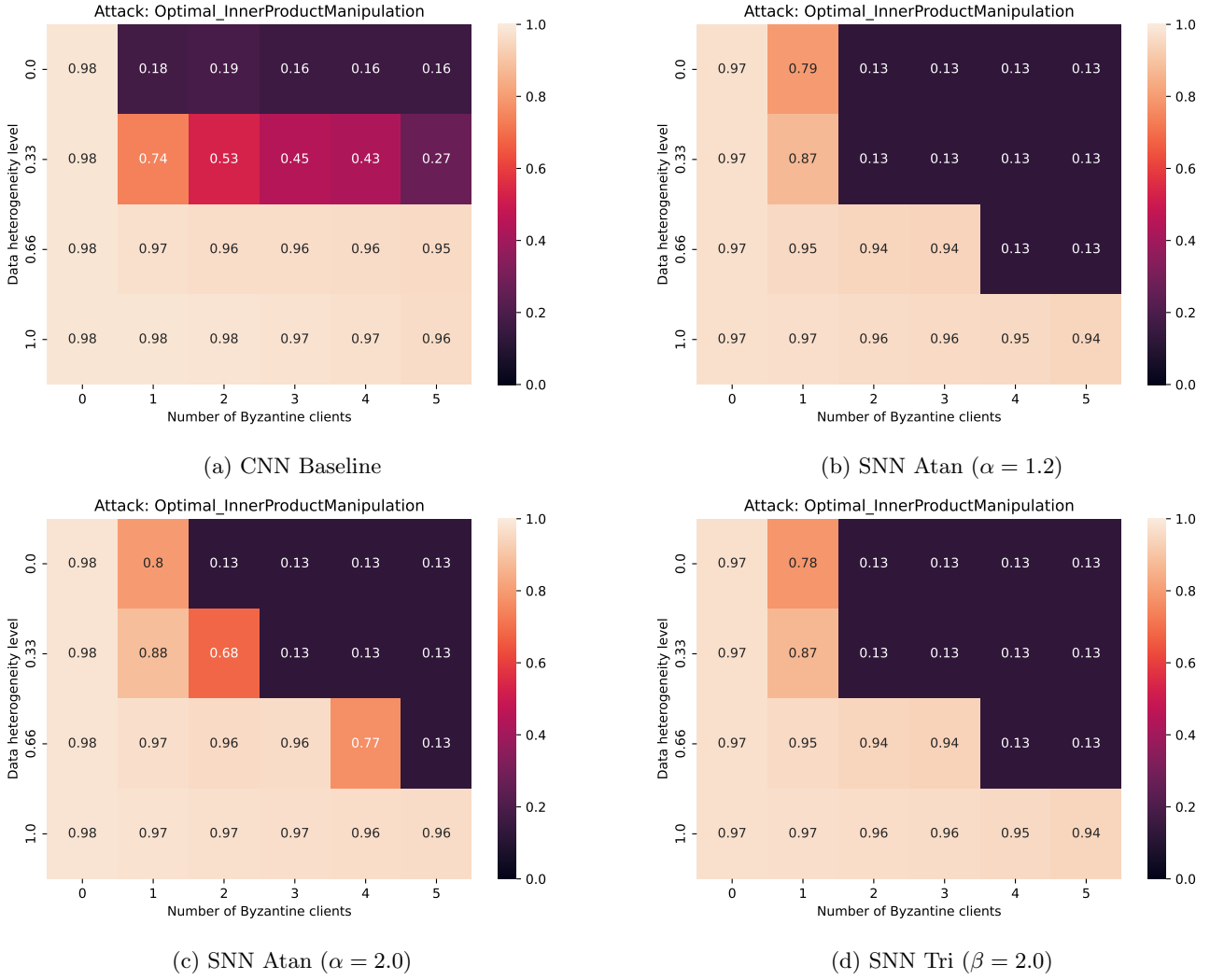
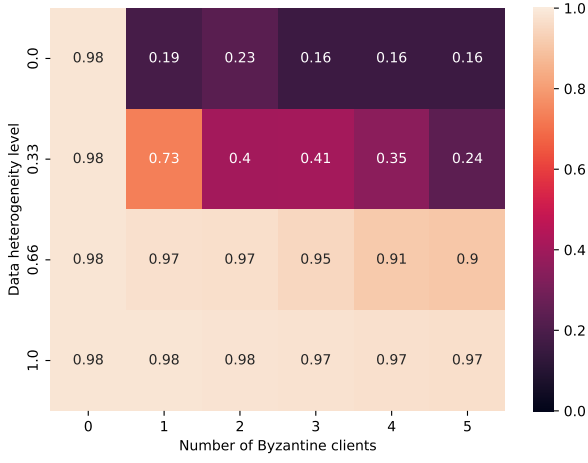


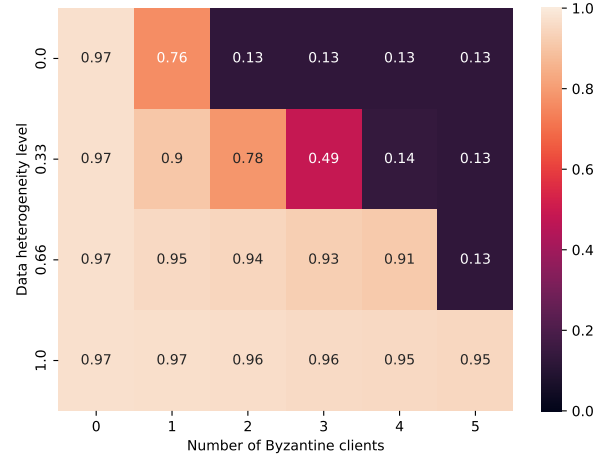
Figure 16: Multi-Krum performance under Optimal Inner Product Manipulation for CNN Baseline vs. SNN Atan ($\alpha = 1.2, 2.0$) vs. SNN Tri ($\beta = 2.0$).

4.4 Trimmed Mean (TM)

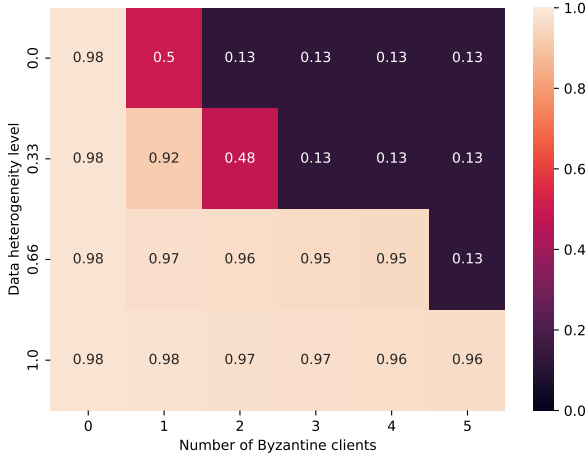
4.4.1 TM under Worst-Case across all Attacks



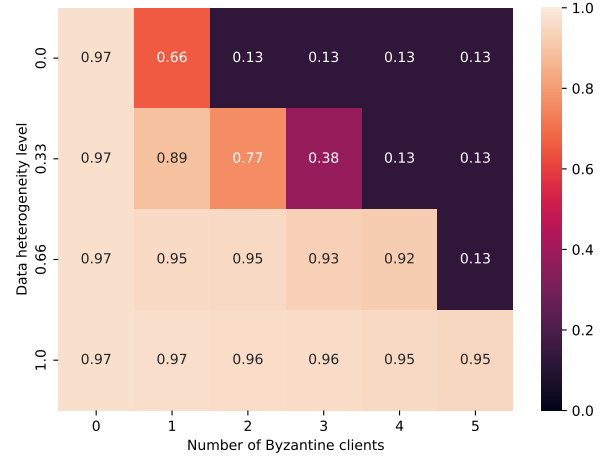
(a) CNN Baseline



(b) SNN Atan ($\alpha = 1.2$)



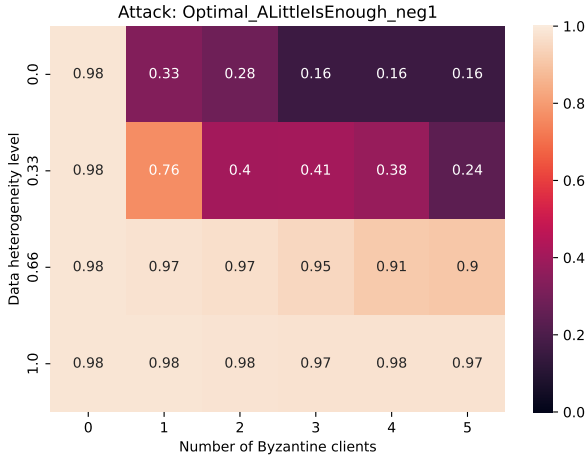
(c) SNN Atan ($\alpha = 2.0$)



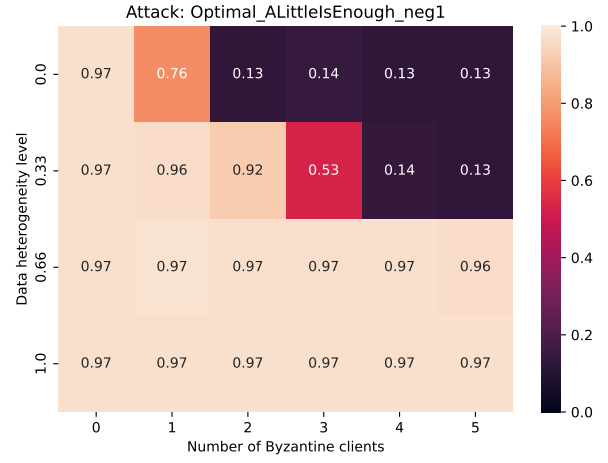
(d) SNN Tri ($\beta = 2.0$)

Figure 17: Trimmed Mean performance under the worst-case across all attacks for CNN Baseline vs. SNN Atan ($\alpha = 1.2, 2.0$) vs. SNN Tri ($\beta = 2.0$).

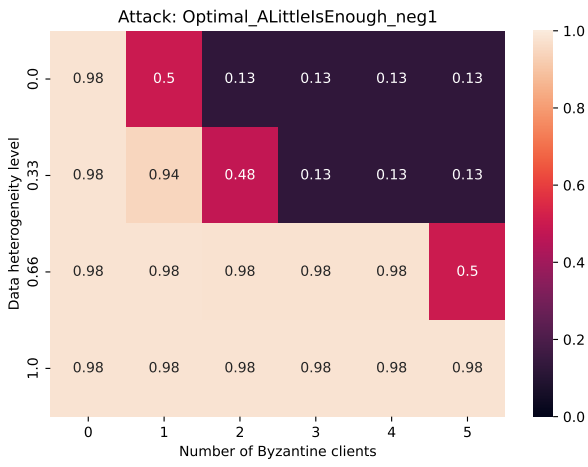
4.4.2 TM under Optimal ALIE Attack



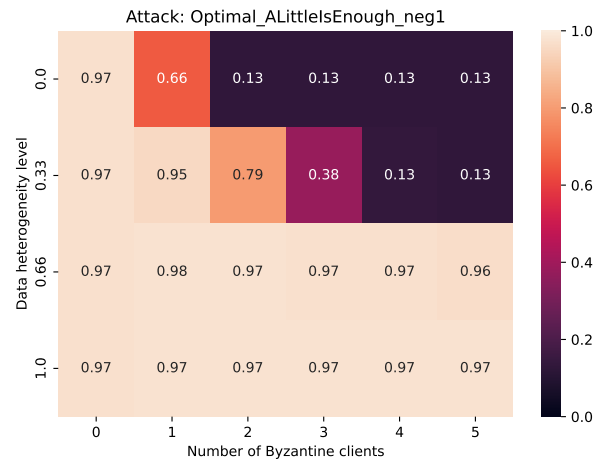
(a) CNN Baseline



(b) SNN Atan ($\alpha = 1.2$)



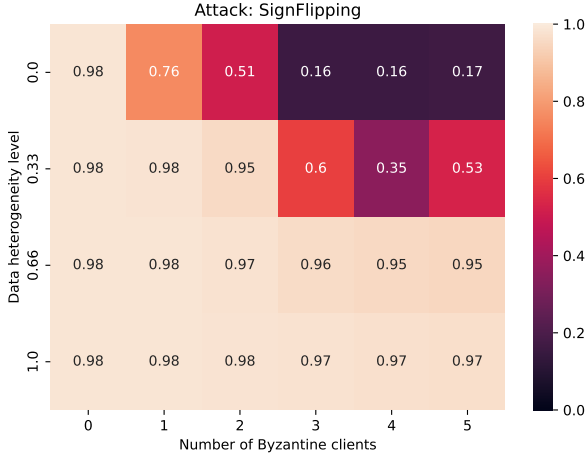
(c) SNN Atan ($\alpha = 2.0$)



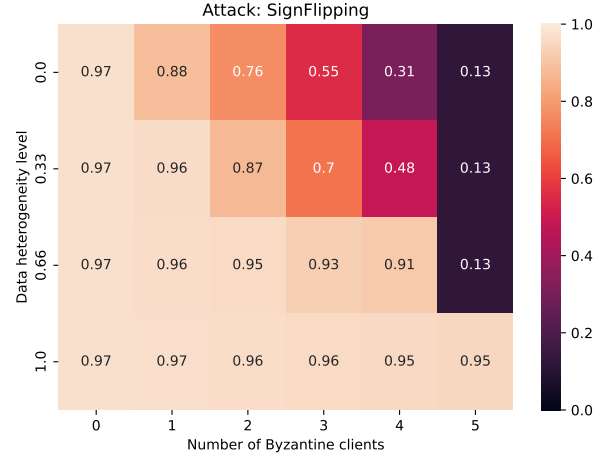
(d) SNN Tri ($\beta = 2.0$)

Figure 18: Trimmed Mean performance under Optimal ALIE for CNN Baseline vs. SNN Atan ($\alpha = 1.2, 2.0$) vs. SNN Tri ($\beta = 2.0$).

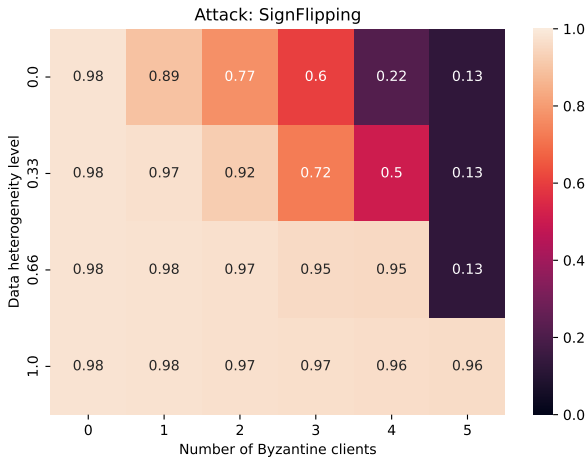
4.4.3 TM under Sign Flipping Attack



(a) CNN Baseline



(b) SNN Atan ($\alpha = 1.2$)



(c) SNN Atan ($\alpha = 2.0$)



(d) SNN Tri ($\beta = 2.0$)

Figure 19: Trimmed Mean performance under Sign Flipping for CNN Baseline vs. SNN Atan ($\alpha = 1.2, 2.0$) vs. SNN Tri ($\beta = 2.0$).

4.4.4 TM under Optimal Inner Product Manipulation Attack

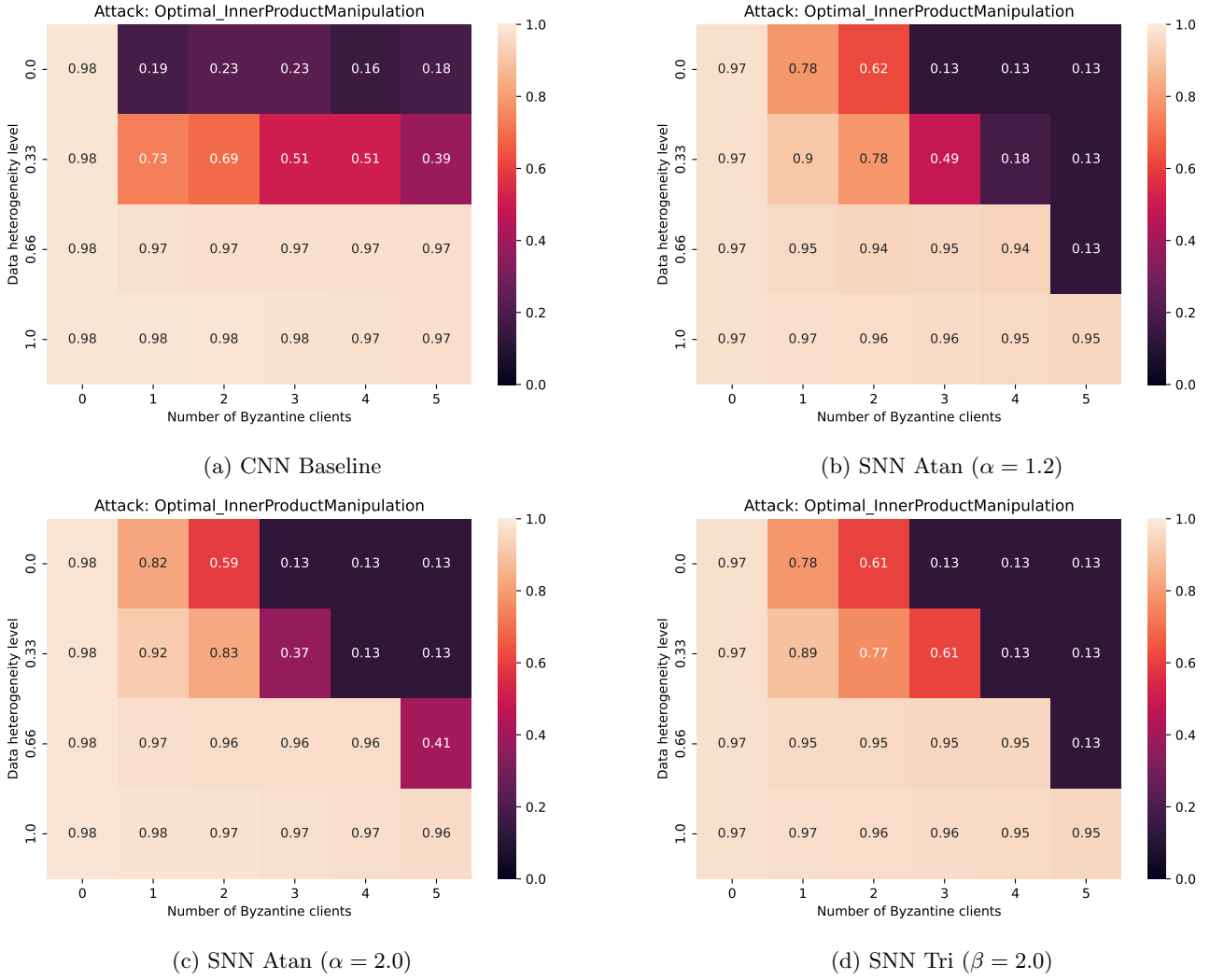


Figure 20: Trimmed Mean performance under Optimal Inner Product Manipulation for CNN Baseline vs. SNN Atan ($\alpha = 1.2, 2.0$) vs. SNN Tri ($\beta = 2.0$).